

20250916EB_MN (vLH51, vLH103, virus titer test)

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Status: In progress

Entry Created On: 01 Oct 2025 20:39:28 UTC

Entry Last Modified: 08 Oct 2025 22:26:24 UTC

Export Generated On: 01 Sep 2026 18:37:05 UTC

2025年9月16日 星期二

split iPSCs into a full 6-well plate on the Friday before differentiation
differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates
keep some iPSCs

2025年9月18日 星期四

T2 -> T3

2025年9月20日 星期六

T3 -> T4

2025年9月22日 星期一

T4 -> T5

2025年9月24日 星期三

T5 -> T6

2025年9月26日 星期五

T6 -> T6 (~14mL for over-weekend)

2025年9月27日 星期六

T6 -> T6 (I got covid, so I dissociate the cells the day after)

2025年9月30日 星期二

T6 -> T7

Dissociation

1. Coat 0.01% Poly-L-Ornithine (PLO) Solution with sodium borate (1:100) over weekend
2. Remove the PLO and wash the plate with PBS once
3. Coat the plate with laminine for 1hr.
4. Collect EBs in 50mL falcon and spin them down with 200rcf, 1min. Remove the supernatant (can leave some liquid to avoid remove EBs).
5. Add 2mL accutase and tap the cells

6. Put the tube into 37C water bath. Flick the tube every 5min until 50min, but the EBs looked very intact. I pipetted it up and down for 30 times, but the EBs were still there. I transfer the supernatant to process the following steps and add 1mL new accutase to remaining EBs.
7. Add 2x volumn of Heat Inactivated FBS into the tube and pipette it with 5mL serological pipette gently.
8. Add 2x volumn of CTWM and pipette it with 5mL serological pipette gently.
9. Filter the cell with 40um filter into a new 50mL tube
10. Spin the cell down at 200rcf, 3.5min. Remove the supernatent. Add 1mL of T7 and pipette it gently and thoroughly.
11. Add 4mL more T7 to resuspend the cells and count it.
12. The yield was too low, so I processed batch 2 after 30min incubation of new 1mL accutase -> 1ml HI-FBS and CTWM -> filter -> centrifuge
13. Plate the cells with T7.

cell info

	A	B	C	D
1	cell	passage	well#	note
2	WT11		6	this batch had been grown on the matrigel with Corning DMEM

Table1

	A	B
1	run1	volume (mL)
2	accutase	2
3	HI-FBS/CTWM	4
4	run2	*take some undissolved EB
5	accutase	1
6	HI-FBS/CTWM	1

Table3

	A	B	C	D	E	F	G	H	I
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	volume
2	4.26	99	7	29.82	A	96	60	36	507
3					B	24	250	16	939
4					C,D	6	1972	12	the rest

Well1							^
	1	2	3	4	5	6	
A				vLH80 1uL	vLH103 1uL	vLH51 (2uL in total)	
B				vLH80 5uL	vLH103 1uL		
C				vLH67 1uL	vLH103 5uL		
D				vLH67 5uL	vLH103 5uL		

2025年10月1日 星期三

Change T7 -> T7

2025年10月3日 星期五

T7 -> BP T8

2025年10月4日 星期六

1/2 T8 -> 1/2 T8

2025年10月7日 星期二

1/2 T8 -> 1/2 T8

Treat C6 and D6

Prepare NB-A with 2.5mM and 0.15mM glucose with full supplement

40mL NB-A + 400uL Glutamax + 400uL N2 + 800uL B27 w/o vitA + 400uL 100mM sodium pyruvate + 40uL T8 supplements

31mL NBA w/ 1mM sodium pyruvate + 310uL Glutamax + 310uL N2 + 620uL B27 w/o vitA + 31uL T8 supplements

2.5mM: 440x from 1.1M glucose (20mL: 45.5uL 1.1M glucose)

0.15mM: 16.67x from 2.5mM glucose or 7333x from 1.1M glucose (2.73uL 1.1M glucose)

2025年10月8日 星期三

Imaging

vLH51

60x, 1.5x zoom -> 0.12um/px

GEM movie 2s only.

51_O.C. , 60X, 1.5xzoom, 600x900, 12-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	488	12	100	10ms		

fileid condition					^
	A	B	C	D	
1	condition#	condition	soma #	neurite #	
2	1	2.5mM glucose 1h	28-40	21-30	
3	2	0.15mM glucose 1h	41-55	31-40	
4	3	2.5mM glucose 24h	1-13	1-10	
5	4	0.15mM glucose 24h	14-27	11-20	

vLH103

60x, 1.5x zoom -> 0.12um/px

Fileid:

- odd: GEM movie, 2s, 200 loops
- even: snapshot

*Note:

- 103-1 was taken twice. The first time I did not turn off the incubator light. The second is the retake of the same positions.
- I should add less 103 next time.

103_O.C. , 60X, 1.5xzoom, 600x900, 12-bit						^
	A	B	C	D	E	
1	channel	bit	power (%)	exposure time	note	
2	488	12	100	10ms		
3	640	12	15	100 ms	snapshot	
4	488	12	20	100 ms		

Table4

	A	B	C	D	E	F
1			vLH51		vLH103	
2	condition#	condition	treat time	image time	treat time	image time
3	1	2.5mM glucose 1h	16:30	17:30	12:30	14:04
4	2	0.15mM glucose 1h	17:04	18:02	13:00	14:17
5	3	2.5mM glucose 24h	10/7 15:47	10/8 16:40-17:00	13:28	14:27
6	4	0.15mM glucose 24h	10/7 15:47	10/8 17:00	14:08	15:07

Well2

	1	2	3	4	5	6
A					1	1
B					2	2
C					1	3
D					2	4

Table2

	A	B
1	condition#	condition
2	1	2.5mM glucose 1h
3	2	0.15mM glucose 1h
4	3	2.5mM glucose 24h
5	4	0.15mM glucose 24h

Table5 ^

	A	B	C	D
1		vLH80	vLH67	
2	2uL	signal is ok and decent (>150 with 12bit). Many TdTomato+ cells	Way too many GEMs. Need to reduce the virus volume added. The signal in neurite is ok. But it's really hard to find ok cells in soma.	
3	5uL	signal is brighter than 2uL (mostly >200 with the same setting). Can't tell much difference regarding the percentage of tdT+ cells.	Way too many GEMs. Even neurites have too many GEMs.	

80_O.C. , 60X,, 600x900, 12-bit ^

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	12	20	200ms	